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COLORADO SCHOOL OF MINES  
MATH 455: Partial Differential Equations

**Assignment #2 - More Fourier Series**  
Due Thursday, September 17, 2026

1. (12 points) Define the function

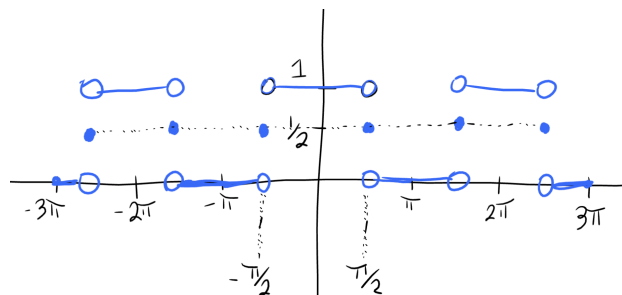
$$f(x) = \begin{cases} 1, & \text{if } |x| < \frac{\pi}{2} \\ 0, & \text{if } |x| > \frac{\pi}{2}, \end{cases}$$

and let  $F(x)$  be the Fourier series of  $f(x)$  on the interval  $(-\pi, \pi)$ . Sketch by hand  $F(x)$  on the interval  $(-3\pi, 3\pi)$  and compute a formula for  $F(x)$  using the Fourier coefficients.

**Solution** — Since  $f$  is not periodic on  $\mathbb{R}$ , we restrict its domain to  $[-\pi, \pi]$  and fill in left-right-averaged discontinuities:

$$f_{2\pi}(x) = \begin{cases} 1 & \text{if } |x| < \frac{\pi}{2} \\ 0 & \text{if } \frac{\pi}{2} < |x| \leq \pi \\ \frac{1}{2} & \text{if } |x| = \frac{\pi}{2} \end{cases}$$

Now let  $g(x)$  be the periodic extension of  $f_{2\pi}(x)$  to all  $\mathbb{R}$ . By the Fourier convergence theorem,  $F(x) = g(x)$  for all  $\mathbb{R}$ , so sketching  $F(x)$  on  $(-3\pi, 3\pi)$  is equivalent to sketching  $g(x)$  on  $(-3\pi, 3\pi)$ :



We compute the Fourier series of  $g(x)$ :

$$\begin{aligned}
a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(x) dx = 1 \\
b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(x) \sin(nx) dx = 0 \quad (\text{by oddness of product } g(x) \sin(nx)) \\
a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} g(x) \cos(nx) dx \\
&= \frac{1}{\pi} \int_{-\pi/2}^{\pi/2} \cos(nx) dx \\
&= \frac{1}{\pi} \left[ \frac{1}{n} \sin(nx) \Big|_{-\pi/2}^{\pi/2} \right] \\
&= \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right)
\end{aligned}$$

Hence

$$\begin{aligned}
F(x) &= \frac{1}{2} + \sum_{n=1}^{\infty} \left[ \frac{2}{n\pi} \sin\left(\frac{n\pi}{2}\right) \right] \cos(nx) \\
&= \frac{1}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \left[ \frac{1}{n} \sin\left(\frac{n\pi}{2}\right) \right] \cos(nx)
\end{aligned}$$

Since

$$\sin\left(\frac{n\pi}{2}\right) = \begin{cases} 0 & \text{if } n \text{ even} \\ 1 & \text{if } n \equiv 1 \pmod{4} \\ -1 & \text{if } n \equiv 3 \pmod{4} \end{cases}$$

we want to skip the even terms, hopping over the odd terms instead, and alternating between a 1 and  $-1$  coefficient, starting with 1:

$$F(x) = \frac{1}{2} + \frac{2}{\pi} \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} \cos((2n+1)x)$$

**2.** (8 points) We know that if  $f(x)$  is odd on the interval  $(-\pi, \pi)$ , then its Fourier series is composed of only sine functions. What additional symmetry condition on  $f$  will make the sine coefficients with even indices become zero (i.e.,  $b_{2n} = 0$  for all  $n \in \mathbb{Z}$ )? Provide a specific example of a function with this property and show that it satisfies the condition.

**Solution** — Let  $f(x)$  be odd. We want to find additional conditions on  $f(x)$  that will force  $b_n = 0$  for even  $n$ , or equivalently, to force  $b_{2n} = 0$  for all  $n$ :

$$\begin{aligned} b_{2n} &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(2nx) dx = 0 \\ &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin(2nx) dx \quad (\text{by evenness of } f(x) \sin(2nx)) \end{aligned}$$

Note that  $\sin(2nx)$  is odd about  $x = \frac{\pi}{2}$ :

$$\begin{aligned} &\sin\left(2n\left(\frac{\pi}{2} + h\right)\right) \\ &= \sin(n\pi + 2nh) \\ &= \sin(n\pi) \cos(2nh) + \cos(n\pi) \sin(2nh) \\ &= -(\sin(n\pi) \cos(2nh) - \cos(n\pi) \sin(2nh)) \\ &= -\sin(n\pi - 2nh) \\ &= -\sin\left(2n\left(\frac{\pi}{2} - h\right)\right) \end{aligned}$$

$x = \frac{\pi}{2}$  also happens to be the midpoint of our integration range  $[0, \pi]$ , so the  $[0, \pi]$ -integral of any function that is odd about  $x = \frac{\pi}{2}$  must be zero. Hence, for  $\int_0^{\pi} f(x) \sin(2nx) dx$  to be zero, it is sufficient for  $f(x)$  to be **even about**  $x = \frac{\pi}{2}$  (in addition to being odd about  $x = 0$ ).

One such function that is odd (about  $x = 0$ ) and even about  $x = \frac{\pi}{2}$  is

$$\boxed{f(x) = \sin(x)}$$

$$\begin{aligned} \sin(h) &= \sin(0 + h) \\ &= \sin(0) \cos(h) + \cos(0) \sin(h) \\ &= -\sin(0) \sin(h) + \cos(0) \sin(h) \\ &= -(\sin(0) \sin(h) - \cos(0) \sin(h)) \\ &= -\sin(0 - h) \\ &= -\sin(-h) \end{aligned}$$

$$\begin{aligned} \sin\left(\frac{\pi}{2} + h\right) &= \sin\left(\frac{\pi}{2}\right) \cos(h) + \cos\left(\frac{\pi}{2}\right) \sin(h) \\ &= \sin\left(\frac{\pi}{2}\right) \cos(h) - \cos\left(\frac{\pi}{2}\right) \sin(h) \\ &= \sin\left(\frac{\pi}{2} - h\right) \end{aligned}$$

3. (12 points) Let

$$f(x) = x \cos(x), \quad x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

$$g(x) = x \cos(x), \quad x \in (-1, 1),$$

and

$$h(x) = \begin{cases} 0, & \text{if } 1 < x < 3 \\ 1, & \text{if } -1 < x < 1 \\ x, & \text{if } -3 < x < -1 \end{cases}$$

be periodic functions (defined via periodic extension) for  $x \in \mathbb{R}$ . For each function

- (a) Determine whether or not the function is in  $PC^1$  on the interval over which it is defined.

**Solution** —  $f(x) = x \cos(x) \in PC^1[-\pi/2, \pi/2]$  because both  $x$  and  $\cos(x)$  are in  $C^\infty(\mathbb{R})$ , so their product must be in  $C^\infty(\mathbb{R})$ , which implies  $f(x) \in PC^1[-\pi/2, \pi/2]$ .

$g(x) = x \cos(x) \in PC^1[-1, 1]$  for an identical reason.

$h(x) \in PC^1[-3, 3]$  because it is built from finitely-many slices of 0, 1, and  $x$ , all three of which are in  $C^\infty(\mathbb{R})$  (which is sufficient to imply that their first derivatives approach finite values at all the junction points).

- (b) State the function to which its Fourier series (defined on the interval:  $[-\frac{\pi}{2}, \frac{\pi}{2}]$  for  $f(x)$ ,  $[-1, 1]$  for  $g(x)$ , and  $[-3, 3]$  for  $h(x)$ ) converges

Note: Do not compute the Fourier series (or coefficients) to do this!

**Solution** — We use the Fourier convergence theorem, which asserts that for a function  $f \in PC^1[a, b]$ , its Fourier series shall converge to the midpoint  $(f(x_0^+) + f(x_0^-))/2$  of the left and right limits at each point  $x = x_0$ .

Since  $f(x) \in PC^1[-\pi/2, \pi/2]$  and  $f(-\pi/2) = f(\pi/2)$  its Fourier series  $F(x)$  is exactly the periodic extension of  $f(x)$  to all  $x \in \mathbb{R}$ :

$$F(x) = x \cos(x) \quad \text{for } x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \text{With } F(x + n\pi) := F(x) \text{ for } n \in \mathbb{Z}$$

Since  $g(x) \in PC^1[-1, 1]$  and  $g(-1) \neq g(1)$ , its Fourier series  $G(x)$  is the periodic extension of  $g(x)$  to all  $x \in \mathbb{R}$  but with left-right-averaged jump discontinuities filled in (left endpoint  $-\cos(1)$  and right endpoint  $\cos(1)$  averaging to 0):

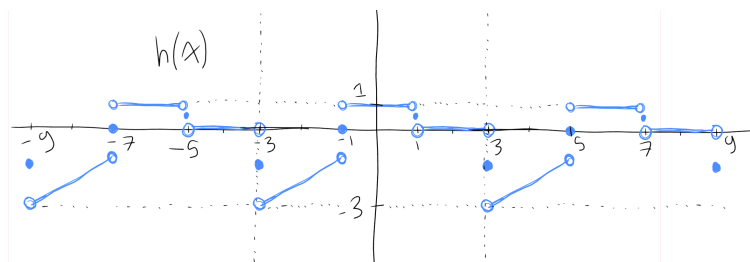
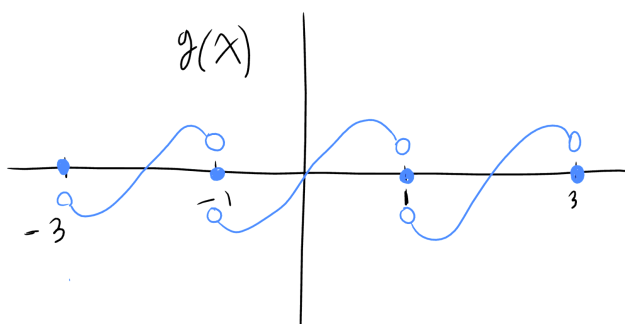
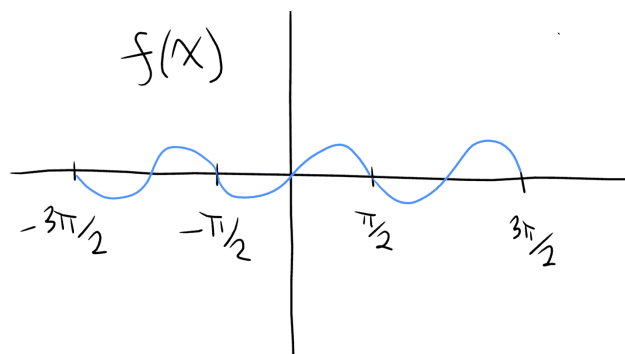
$$G(x) = \begin{cases} x \cos(x), & \text{if } |x| < 1 \\ 0, & \text{if } |x| = 1 \end{cases} \quad \text{With } G(x + 2n) := G(x) \text{ for } n \in \mathbb{Z}$$

Similarly, since  $h(x) \in PC^1[-3, 3]$  the Fourier series of  $h(x)$  converges to the periodic extension of  $h(x)$  with left-right-averaged jump discontinuities filled in:

$$H(x) = \begin{cases} 0, & \text{if } 1 < x < 3 \\ \frac{1}{2}, & \text{if } x = 1 \\ 1, & \text{if } -1 < x < 1 \\ 0, & \text{if } x = -1 \\ x, & \text{if } -3 < x < -1 \\ -\frac{3}{2}, & \text{if } |x| = 3 \end{cases} \quad \text{With } H(x + 6n) := H(x) \text{ for } n \in \mathbb{Z}$$

(c) Sketch by hand three periods of the Fourier series in part (b).

**Solution —**



4. (8 points) Let

$$F(x) = \sum_{n=1}^{\infty} \frac{(-1)^n n^{3/2} + 2}{n^8 + 2n^6 + 1} \cos(nx)$$

How many continuous derivatives can we be sure that  $F$  possesses? Justify your answer using tools from Calculus II.

**Solution** — For an infinite series-representation of a function, it is guaranteed to be continuous if the series converges uniformly for all  $x$ . Hence we must find how many  $x$ -derivatives we can apply to  $F$  and still have a uniformly-convergent series. Let

$$a_n = \frac{(-1)^n n^{3/2} + 2}{n^8 + 2n^6 + 1}$$

Since  $|\cos(nx)| \leq 1$  (also regardless of  $x$ ), we have  $|a_n \cos(nx)| \leq |a_n|$ , so each term is bounded in magnitude by  $|a_n|$ . Hence  $F$  converges independently of  $x$  (and uniformly), because  $\sum |a_n|$  behaves like a convergent  $p$ -series with asymptotic term behavior:

$$|a_n| \sim \frac{1}{n^{13/2}} \quad \text{as } n \rightarrow \infty$$

After differentiating  $F(x)$   $k$  times with respect to  $x$ , the  $n$ th term has asymptotic magnitude

$$n^k |a_n| \sim \frac{1}{n^{13/2-k}}$$

We see that we exit the realm of comfortably (uniformly) converging series at  $k = 6$  derivatives, where the term coefficient  $n^k |a_n| \sim n^{-1/2}$ . Because  $F^{(6)}(x)$  behaves as a reciprocal-of-square-roots series, it is not guaranteed to converge for all  $x$  (much less uniformly). Hence,  $F$  is guaranteed to have  $\boxed{5}$  continuous derivatives.